Chapter 10 solutions

CH10-E1

1. Use -1 at the tail and $+1$ at the head.

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

2. The graph is connected, so

$$\operatorname{rank} A = 4 - 1 = 3, \qquad \dim N(A) = 1.$$

The kernel is the span of $(1, 1, 1, 1)^T$.

3. With $m = n = 4, r = 3$, left nullity is 1. The cycle $1 \to 2 \to 3 \to 1$ has edge signs $+1, +1, -1$, giving

$$y = (1, 1, -1, 0)^T, \qquad A^T y = 0.$$

The final edge $3 \to 4$ is outside the cycle.

4. Choose $x = (0, 0, 0, 2)^T$, whose differences are $(0, 0, 0, 2)^T$. Pairing with the cycle flow gives

$$y^T A x = 0 + 0 + (-1)0 + 0 = 0.$$

This verifies orthogonality between potential differences and circulations.

CH10-E2

1. Delete the row for edge $2 \to 3$.

$$B = \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

$$L = B^T B = \begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

2. Since $B\mathbf{1} = 0$, we have $L\mathbf{1} = 0$. Symmetry gives zero column sums as well.

3. Delete row and column 4 to ground node 4.

$$L_{\text{red}} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 2 \end{bmatrix}.$$

4. Solve

$$2x_1 - x_2 - x_3 = 1,$$

Chapter 10 solutions (continued)

$$-x_1 + x_2 = 0, \qquad -x_1 + 2x_3 = 0.$$

The last two equations give $x_2 = x_1$ and $x_3 = x_1/2$. The first then gives $x_1 = 2$, so

$$x = (2, 2, 1, 0)^T.$$

Edge differences:

$$Bx = (0, -1, -1)^T.$$

Substitution gives

$$B^T Bx = (1, 0, 0, -1)^T,$$

The first three entries match the reduced equation. The final -1 balances flow at the ground node. The complete Laplacian right-hand side must sum to zero.

CH10-E3

1. With fixed ends, the unknown displacements are u_1, u_2, u_3.

$$e = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}.$$

2. For $C = \mathrm{diag}(c_1, c_2, c_3, c_4)$,

$$K = A^T CA = \begin{bmatrix} c_1 + c_2 & -c_2 & 0 \\ -c_2 & c_2 + c_3 & -c_3 \\ 0 & -c_3 & c_3 + c_4 \end{bmatrix}.$$

3. If every $c_i = 1$,

$$K = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

for nonzero u,

$$u^T Ku = \sum_i e_i^2 > 0.$$

the fixed boundaries and connected springs force $Au = 0$ only at $u = 0$, so K is positive definite. Leading minors $2, 3, 4$ provide another check.

4. For the same three masses, free ends remove the two springs attached to walls. Set $c_1 = c_4 = 0$, retaining $c_2, c_3 > 0$. Then

$$u^T Ku = c_2(u_2 - u_1)^2 + c_3(u_3 - u_2)^2.$$

zero energy holds exactly when $u_1 = u_2 = u_3$, so $N(K) = \mathrm{span}\{(1, 1, 1)^T\}$. Translating all masses together stretches no spring and produces no restoring force.

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Chapter 10 solutions (continued)

CH10-E4

1. Nonnegative entries and unit column sums prove the stated convention.
2. $$\det(P-\lambda I)=(0.8-\lambda)(0.7-\lambda)-0.06=\lambda^2-1.5\lambda+0.5.$$

The eigenvalues are 1 and 0.5. For $\lambda=1$,

$$(P-I)v=0$$

writing a generic eigenvector as $(a,b)^T$ gives $2a=3b$. Choose stationary eigenvector $(3,2)^T$. For $\lambda=0.5$, choose

$$v_2=(1,-1)^T.$$

3. Normalize the stationary eigenvector to total 1.

$$u_\infty=(0.6,0.4)^T.$$

4. Expand

$$u_0=au_\infty+b(1,-1)^T.$$

The second component gives $b=0.4a$, and the first gives

$$1=0.6a+b=0.6a+0.4a=a.$$

Hence $a=1, b=0.4$, so

$$u_k=u_\infty+0.4(0.5)^k(1,-1)^T.$$

The nonstationary factor 0.5^k gives the exact convergence rate in this example.

CH10-E5

1. $$\det(A-\lambda I)=(0.2-\lambda)(0.1-\lambda)-0.12=\lambda^2-0.3\lambda-0.1.$$

Roots:

$$\lambda=\frac{0.3\pm\sqrt{0.09+0.4}}{2}=\frac{0.3\pm0.7}{2},$$

The roots are $0.5, -0.2$, so the spectral radius is 0.5.

2. $$I-A=\begin{bmatrix}0.8 & -0.3\\ -0.4 & 0.9\end{bmatrix}.$$

The determinant is $0.72-0.12=0.6$, giving

$$(I-A)^{-1}=\frac{1}{0.6}\begin{bmatrix}0.9 & 0.3\\ 0.4 & 0.8\end{bmatrix}.$$

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Chapter 10 solutions (continued)

3.

$$p = (I - A)^{-1} \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

The components are $(0.9 + 0.6)/0.6 = 2.5$ and $(0.4 + 1.6)/0.6 = 10/3$. Therefore

$$p = \left(\frac{5}{2}, \frac{10}{3} \right)^T.$$

4. The spectral radius is below 1, so powers tend to zero and the Neumann series converges absolutely. Nonnegativity makes its terms direct demand, indirect demand, and successive rounds of intermediate demand.

CH10-E6

1. The equality constraint is

$$x_1 + x_2 + 2x_3 = 4.$$

Set two nonbasic variables to zero to obtain vertices

$$(4, 0, 0), \quad (0, 4, 0), \quad (0, 0, 2).$$

Their convex hull is the complete feasible triangle.

2. Objective values:

$$20, \quad 12, \quad 16.$$

The unique minimum is 12 at $(0, 4, 0)$.

3. Primal:

$$\min 5x_1 + 3x_2 + 8x_3$$

subject to $x_1 + x_2 + 2x_3 = 4, x \geq 0$.

Dual:

$$\max 4y$$

subject to

$$y \leq 5, \quad y \leq 3, \quad 2y \leq 8.$$

The dual point $y = 3$ is feasible and gives value 12. Weak duality then proves optimality with zero gap.

4. The three vertex costs differ strictly, giving one optimal vertex and hence one optimum on this triangle. Distinct reported optima require checking tolerances or errors. In a general LP, if two distinct vertices are optimal, their joining segment is optimal. Different bases representing one point instead indicate degeneracy.

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Chapter 10 solutions (continued)

CH10-E7

1. The mean over one period is zero. Also $f(2\pi - x) = -f(x)$ away from the jump endpoints, while cosine is unchanged, so the cosine integrals cancel.
2. $$b_n = \frac{1}{\pi}\left(\int_0^{\pi} \sin(nx)\,dx - \int_{\pi}^{2\pi} \sin(nx)\,dx\right).$$

This gives

$$b_n = \begin{cases} \dfrac{4}{\pi n}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

3. $$f_N(x) = \frac{4}{\pi} \sum_{\substack{1 \le n \le N \\ n \text{ odd}}} \frac{\sin(nx)}{n}.$$
4. Each coefficient multiplies the orthogonal projection onto its trigonometric direction. Truncation omits the higher orthogonal components.

$$\|f - f_N\|_2^2 = \pi \sum_{n > N, n \text{ odd}} \frac{16}{\pi^2 n^2} = \frac{16}{\pi} \sum_{n > N, n \text{ odd}} \frac{1}{n^2}.$$

Since $\|\sin nx\|_2^2 = \pi$, the basis-norm factor must be retained. Adding a nonzero odd mode strictly decreases error; adding a zero-coefficient even mode leaves it unchanged. The finite identity $\|f - f_N\|^2 = 2\pi - \pi \sum_{n \le N} b_n^2$ checks the result without first using infinite Parseval. Pointwise behavior near jumps involves Gibbs oscillations and is a separate question.

CH10-E8

1. Use homogeneous column coordinates.

$$T(2, -1) = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix},$$

$$S(3) = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$R_{90} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Chapter 10 solutions (continued)

2. Rotation about $c = (2, -1)$ gives

$$M = T(c)R_{90}T(-c) = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -3 \\ 0 & 0 & 1 \end{bmatrix}.$$

Define the homogeneous center $c_h = (2, -1, 1)^T$. Direct multiplication gives $Mc_h = c_h$. Multiplication of the 3×3 matrix by the two-dimensional c would be undefined.

3. The point $(1, 0, 1)^T$ becomes $(3, -1, 1)^T$. The direction $(1, 0, 0)^T$ is unchanged because its last coordinate is zero.

4. Unit normal $n = (1, 2)^T/\sqrt{5}$:

$$P = I - nn^T = \begin{bmatrix} 4/5 & -2/5 \\ -2/5 & 1/5 \end{bmatrix},$$

$$M = I - 2nn^T = \begin{bmatrix} 3/5 & -4/5 \\ -4/5 & -3/5 \end{bmatrix}.$$

P projects onto the line $n^T x = 0$ through the origin, and M reflects across that line. For column vectors, the rightmost matrix acts first. Since $x = Px + (n^T x)n$, reflection gives $Mx = Px - (n^T x)n = 2Px - x$.

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CH10-E9

1.

$$\det K = 1 \cdot 5 - 2 \cdot 3 = -1 \equiv 28 \pmod{29}.$$

Since 29 is prime and 28 is nonzero modulo 29, the key is invertible.

2. 2×2 inverse:

$$K^{-1} = 28^{-1} \begin{bmatrix} 5 & -2 \\ -3 & 1 \end{bmatrix}.$$

The inverse of $28 \equiv -1$ is itself. Negating the adjugate entries gives

$$K^{-1} = \begin{bmatrix} 24 & 2 \\ 3 & 28 \end{bmatrix} \pmod{29}.$$

Check $K^{-1}K = I \pmod{29}$.

3. Message $x = (5, 8)^T$:

$$y_1 = 5 + 16 = 21,$$

$$y_2 = 15 + 40 = 55 \equiv 26 \pmod{29}.$$

The ciphertext block is $(21, 26)^T$.

4. Decode:

Chapter 10 solutions (continued)

$$x_1 = 24(21) + 2(26) \pmod{29} = 5,$$
$$x_2 = 3(21) + 28(26) \pmod{29} = 8.$$

This cipher uses $y = Kx \pmod{29}$, with no translation term. If two known plaintext blocks form invertible $X = [x^{(1)}\ x^{(2)}]$, with corresponding ciphertexts $Y = [y^{(1)}\ y^{(2)}]$, then $Y = KX$ gives $K = YX^{-1} \pmod{29}$. Blocks on a single line may fail to identify K uniquely; their rank matters, not merely their count. This linear structure and small key space do not provide modern cryptographic security.

CH10-E10

1. Using head potential minus tail potential,

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix}, \qquad C = \operatorname{diag}(1, 2, 3, 4).$$

take x in V and c_e in S, or A/V. Then $e = Ax$ is in V; $y = Ce$, $j = -y$, and f are in A. Positive injection adds current to a node, balanced by net outward edge current. Summing the edge outer products gives

$$L = A^T C A = \begin{bmatrix} 3 & -1 & -2 & 0 \\ -1 & 4 & 0 & -3 \\ -2 & 0 & 6 & -4 \\ 0 & -3 & -4 & 7 \end{bmatrix}.$$

2. Reversing the second edge gives

$$D = \operatorname{diag}(1, -1, 1, 1), \qquad A' = DA.$$

Since both D and C are diagonal,

$$(A')^T C A' = A^T\, D\, C\, D\, A = A^T C A = L.$$

the second difference and weighted difference change sign, $e_2' = -e_2$, $y_2' = -y_2$. Therefore $(A')^T y' = A^T\, D^2 y = A^T y = f$. The orientation changes the coordinates, not the physical flow or balance.

3. The injection sum is

$$-4 - 4 - 10 + 18 = 0.$$

Deleting node 1 gives

$$L_{\text{red}} = \begin{bmatrix} 4 & 0 & -3 \\ 0 & 6 & -4 \\ -3 & -4 & 7 \end{bmatrix}, \qquad f_{\text{red}} = (-4, -10, 18)^T.$$

Solve

Chapter 10 solutions (continued)

$$4x_2 - 3x_4 = -4, \qquad 6x_3 - 4x_4 = -10, \qquad -3x_2 - 4x_3 + 7x_4 = 18$$

so $x_2 = 2$, $x_3 = 1$, $x_4 = 4$, and $x = (0, 2, 1, 4)^T$.

4. The differences and weighted differences are

$$e = Ax = (2, 1, 2, 3)^T, \qquad y = Ce = (2, 2, 6, 12)^T.$$

The physical currents are $j = -y = (-2, -2, -6, -12)^T$. Also

$$-A^T j = A^T y = (-4, -4, -10, 18)^T = f.$$

The quadratic functional is

$$\frac{1}{2}x^T Lx = \frac{1}{2}(1 \cdot 2^2 + 2 \cdot 1^2 + 3 \cdot 2^2 + 4 \cdot 3^2) = 27,$$

and

$$\frac{1}{2}f^T x = \frac{1}{2}(-4 \cdot 2 - 10 \cdot 1 + 18 \cdot 4) = 27.$$

the two forms agree and are independent of a common potential offset. Here $x^T Lx = \sum_e c_e e_e^2 = 54$ has units W and is total dissipated power. Its half, 27, is half the power, not stored energy in J. In the spring model, $\frac{1}{2}u^T Ku$ is elastic potential energy. The algebraic forms agree but the physical units differ.

5. Replacing the last source by 17 gives $\mathbf{1}^T f = -1 \neq 0$. Since $A\mathbf{1} = 0$, $\mathbf{1}^T Lx = 0$, so $f \notin C(L) = C(A^T) = N(A)^\perp$ and the full system is inconsistent. A grounded solve can still return a vector, but implicitly changes node 1's injection from -4 to -3, adding $+1$ unit of current. Without an explicitly modeled ground reaction or external source, that silently changes conservation. Correct the data or explicitly add the compensating source and boundary conditions before solving.

Chapter 10 solutions (continued)

CH10-E11

1. Initially $x_1 = 0, s_1 = 0, s_2 = 1, B = I_2$. The reduced cost is -1, the direction is $B^{-1}a_1 = (1, 1)^T$, and the ratios are 0 and 1. Choose 0. Variable x_1 enters and s_1 leaves, giving $x_1 = -s_1, s_2 = 1 + s_1$. The vertex stays fixed.
2. Choosing the strictly positive ratio 1 gives $(x_1, s_1, s_2) = (1, -1, 0)$. Both equalities hold, but $s_1 < 0$ violates feasibility. Even a zero basic value must constrain the step.
3. The new objective is $-x_1 = s_1 \geq 0$, attained at $(0, 0, 1)$, so the optimum is 0. This degenerate basic feasible solution has a zero basic value; a zero-step pivot changes the basis without strictly decreasing the objective.
4. Every feasible point satisfies $c^T x = c_B^T B^{-1} b + \sum_{j \notin B} \bar{c}_j x_j$. Nonnegative reduced costs make this no smaller than the current basic objective. If some $\bar{c}_j < 0$ and $d = B^{-1}a_j \leq 0$, set $x_j = t$, $x_B(t) = B^{-1}b - dt \geq 0$ for any $t \geq 0$. The equalities remain valid while the objective tends to $-\infty$ through $\bar{c}_j t$, certifying unboundedness.

Verification code appendix: applications.py

```
"""Check representative calculations across the chapter ten applications."""

from itertools import product
from math import cos, isclose, pi, sin, sqrt

Mat = tuple[tuple[float, ...], ...]
Vec = tuple[float, ...]


def transpose(a: Mat) -> Mat:
    return tuple(zip(*a))


def matmul(a: Mat, b: Mat) -> Mat:
    return tuple(
        tuple(sum(a[i][k] * b[k][j] for k in range(len(b))) for j in range(len(b[0])))
        for i in range(len(a))
    )


def matvec(a: Mat, x: Vec) -> Vec:
    return tuple(sum(a[i][j] * x[j] for j in range(len(x))) for i in range(len(a)))


def clean(a: Mat) -> Mat:
    return tuple(tuple(0.0 if abs(x) < 1e-12 else x for x in row) for row in a)
```

Chapter 10 solutions (continued)

```
incidence = (
    (-1.0, 1.0, 0.0, 0.0),
    (0.0, -1.0, 1.0, 0.0),
    (-1.0, 0.0, 1.0, 0.0),
    (0.0, 0.0, -1.0, 1.0),
)
laplacian = matmul(transpose(incidence), incidence)
assert matvec(laplacian, (1.0, 1.0, 1.0, 1.0)) == (0.0, 0.0, 0.0, 0.0)
tree_incidence = (incidence[0], incidence[2], incidence[3])
tree_flow = matvec(tree_incidence, (2.0, 2.0, 1.0, 0.0))
assert tree_flow == (0.0, -1.0, -1.0)
assert matvec(transpose(tree_incidence), tree_flow) == (1.0, 0.0, 0.0, -1.0)

spring_difference = (
    (1.0, 0.0, 0.0),
    (-1.0, 1.0, 0.0),
    (0.0, -1.0, 1.0),
    (0.0, 0.0, -1.0),
)
stiffness = matmul(transpose(spring_difference), spring_difference)
assert stiffness == ((2.0, -1.0, 0.0), (-1.0, 2.0, -1.0), (0.0, -1.0, 2.0))

markov = ((0.8, 0.3), (0.2, 0.7))
stationary = (0.6, 0.4)
stationary_image = matvec(markov, stationary)
assert all(isclose(stationary_image[i], stationary[i], abs_tol=1e-12) for i in range(2))
assert abs(sum(row[0] for row in markov) - 1.0) < 1e-12
```

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Chapter 10 solutions (continued)

```
assert abs(sum(row[1] for row in markov) - 1.0) < 1e-12
assert matvec(markov, (1.0, 1.0)) != (1.0, 1.0)

# Enumerate vertices of x1 + x2 + 2x3 = 4 with two variables zero.
lp_candidates = []
for i, j, k in ((0, 1, 2),):
    lp_candidates = [(4.0, 0.0, 0.0), (0.0, 4.0, 0.0), (0.0, 0.0, 2.0)]
lp_cost = tuple(5.0 * x + 3.0 * y + 8.0 * z for x, y, z in lp_candidates)
assert min(lp_cost) == 12.0

graphics_translation = ((1.0, 0.0, 2.0), (0.0, 1.0, -1.0), (0.0, 0.0, 1.0))
graphics_rotation = ((0.0, -1.0, 0.0), (1.0, 0.0, 0.0), (0.0, 0.0, 1.0))
center = (2.0, -1.0, 1.0)
translate_negative = ((1.0, 0.0, -2.0), (0.0, 1.0, 1.0), (0.0, 0.0, 1.0))
rotate_about_center = matmul(matmul(graphics_translation, graphics_rotation),
translate_negative)
assert matvec(rotate_about_center, center) == center

# Hill cipher over F_29.
det_k = 1 * 5 - 2 * 3
assert det_k % 29 != 0
key_inverse = ((24, 2), (3, 28))
cipher = tuple(
    sum(((1, 2), (3, 5))[i][j] * (5, 8)[j] for j in range(2)) % 29
    for i in range(2)
)
assert cipher == (21, 26)
plain = tuple(sum(key_inverse[i][j] * cipher[j] for j in range(2)) % 29 for i in range(2))
assert plain == (5, 8)
```

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Chapter 10 solutions (continued)

```
# Finite Fourier projection for the square wave's first odd modes.
def coefficient(n: int) -> float:
    values = []
    for k in range(6000):
        x = 2.0 * pi * (k + 0.5) / 6000
        value = 1.0 if x < pi else -1.0
        values.append(value * sin(n * x))
    return 2.0 * sum(values) / 6000


assert isclose(coefficient(1), 4.0 / pi, rel_tol=1e-4, abs_tol=1e-4)
assert abs(coefficient(2)) < 1e-4
fourier_error = sum(
    ((1.0 if 2 * pi * (k + 0.5) / 6000 < pi else -1.0)
     - (4 / pi) * sin(2 * pi * (k + 0.5) / 6000)) ** 2
    for k in range(6000)
) * 2 * pi / 6000
assert isclose(fourier_error, 2 * pi - 16 / pi, abs_tol=1e-5)

print("Laplacian =", laplacian)
print("Markov stationary distribution =", stationary)
print("LP minimum =", min(lp_cost))
print("Hill plaintext =", plain)
print("First Fourier coefficient =", coefficient(1))
```

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References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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